

RESEARCH ARTICLE

Physics-Informed Agent-Analytics Framework With Causal Validation for Crease Prediction in Multilayer Flexible Cover Stacks

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ABSTRACT

We propose an agent-analytics framework for data-driven crease mitigation in multilayer flexible cover stacks, where an LLM-assisted workflow supports a human domain expert in formalizing engineering hypotheses into mechanics-grounded descriptors and in organizing ensemble modeling, explainability, and adjustment-based causal-consistency analysis pipelines. Starting from neutral-axis mechanics and the parallel axis theorem, the framework systematically generates 14 physics-informed descriptors organized into five mechanism groups: geometry, stiffness balance, elastic energy, viscoelasticity, and interface mechanics. Through a three-stage evidence hierarchy—ensemble SHAP importance, bootstrap pathway analysis under progressive adjustment, and progressive confounder adjustment—we identify three independently robust descriptors while demonstrating that the remaining descriptors operate through these base mechanisms. Evaluated on 63 physically fabricated stack variants, the 14-model ensemble achieves a best cross-validated $R^2=0.71$. The progression from ratio-based stiffness symmetry (Tier 3) to absolute moment imbalance (Tier 1) is validated by both improved prediction and evidence-tier promotion, confirming a shared stiffness-balance mechanism. A comparative SHAP waterfall analysis reveals that the best-performing deployable design achieves 39% crease reduction below the dataset mean through geometric compensation rather than extreme material reduction, providing actionable prescriptive guidance for stack design.

1 | Introduction

As the form factor of mobile devices evolves toward foldable and rollable designs, the mechanical durability and optical quality of the multilayer cover stack have become paramount. A persistent challenge is the “crease”—a residual surface deformation at the folding area caused by the interplay of long-term viscoelastic creep, plastic deformation of adhesive layers, and localized stress concentration [1–3]. This phenomenon is particularly exacerbated in heterogeneous stacks comprising materials with vastly different moduli, such as ultra-thin glass (rigid cover layer), optical polymers, and pressure-sensitive adhesives (soft interlayers).

Traditionally, finite element analysis (FEA) has been the gold standard for predicting such deformations. However, accurate FEA requires precise nonlinear visco-hyperelastic constitutive models, which are often unavailable for novel developmental materials. Conventional machine learning (ML) can predict outcomes rapidly but typically treats the multilayer stack as a “black box,” relying on raw geometric parameters without capturing the underlying mechanics, thus failing to explain why a specific stack performs well.

To bridge this gap, we propose a physics-informed agent-analytics framework (Figure 1). Rather than treating the LLM solely as a passive text generator, our workflow uses the agent

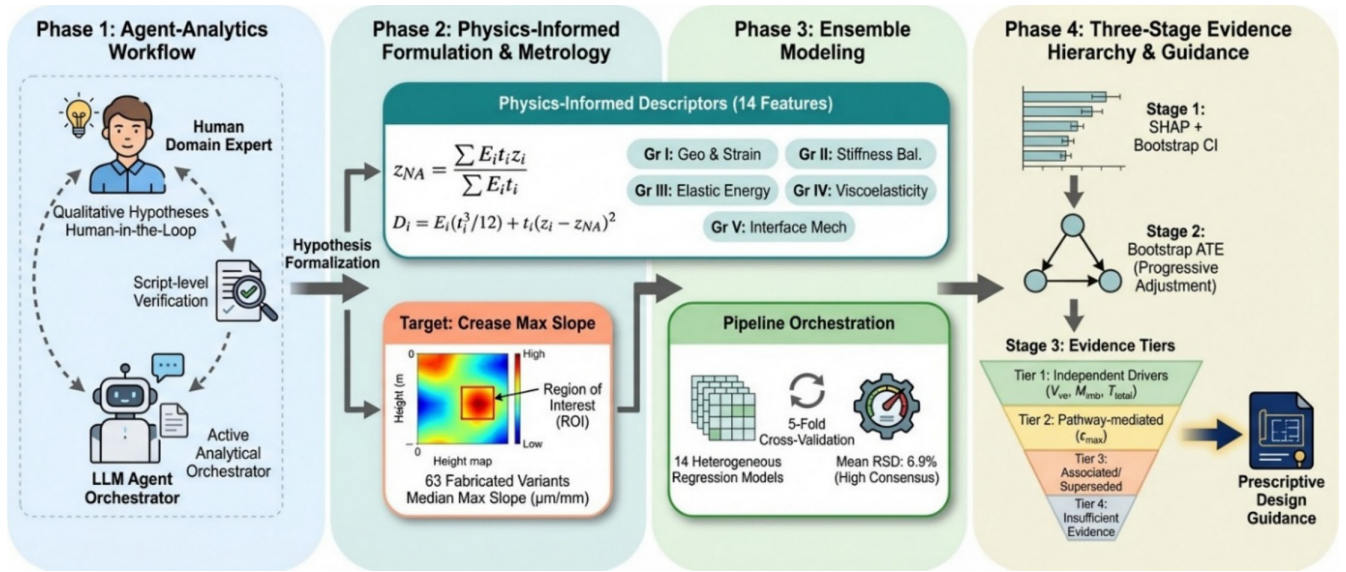


FIGURE 1 | Agent-analytics workflow integrating hypothesis formalization, physics-informed descriptor extraction, ensemble modeling, three-stage evidence hierarchy, and prescriptive design guidance.

as a structured analytical assistant for hypothesis formalization and descriptor organization. In this collaborative workflow, a human domain expert defines qualitative engineering hypotheses, and the agent assists in translating them into mechanics-grounded descriptors. The agent-supported pipeline extends beyond feature engineering to organize downstream evidence quantification, including ensemble ML, SHAP-based explainability, and pathway analysis under progressive adjustment. The human author retains full engineering control, validating all equations and tests to ensure industrial applicability and confidentiality.

We make five contributions:

1. We propose 14 physics-informed descriptors organized into five mechanism groups (geometry, stiffness balance, elastic energy, viscoelasticity, and interface), grounded in neutral-axis mechanics and the parallel axis theorem.
2. We introduce a three-stage evidence hierarchy (ensemble SHAP with bootstrap CI, progressive adjustment-pathway decomposition, and progressive confounder adjustment) that systematically winnows 14 candidates to 3 independently robust descriptors.
3. We identify viscoelastic accumulation (D9), dynamic stiffness imbalance (D5), and total thickness (D1) as the three independently robust crease drivers under minimal sufficient adjustment.
4. We provide convergent adjustment-based evidence that the evolution from ratio-based S_{sym} to absolute M_{imb} represents a qualitative shift in evidence-tier status (Tier 3 $\rightarrow$ Tier 1), not merely a predictive improvement.
5. We demonstrate through quantitative SHAP waterfall analysis that geometric compensation via stiffness balance optimization offers a more holistically viable crease mitigation strategy than material-only approaches.

2 | Related Work

2.1 | Stack Compliance and Viscoelastic Mechanisms

Many crease mitigation studies converge on a common, materials-agnostic mechanism: Residual crease formation is governed by time-dependent interlayer response (stress relaxation/creep) coupled with geometric lever-arm effects. Reported design strategies focus on balancing stiffness above and below the neutral axis, limiting excessive elastic energy storage, and reducing viscoelastic accumulation under repeated bending [4–8]. Because absolute formulations and process recipes are typically proprietary, an industrially viable reporting approach is to disclose only derived indices—the strategy adopted in our framework.

2.2 | Computational Mechanics and Structure Optimization

High-fidelity FEA remains the dominant tool for understanding crease formation and recovery in multilayer foldable stacks [4]. Practical simulation acceleration strategies are frequently used for large-deformation and contact problems, and visco-hyperelastic constitutive modeling is commonly highlighted for reproducing realistic cyclic behavior. However, these simulation-centric approaches require accurate constitutive data that are often unavailable for novel developmental materials—a gap our measurement-anchored framework addresses.

2.3 | AI-Enabled Prediction and Metrology

Surrogate modeling has been adopted to approximate expensive FEA with ML [9–11], and machine-vision approaches demonstrate practical pipelines for defect detection [12, 13], and early-warning methodologies based on critical-angle

curvature analysis have been proposed for preemptive crack vulnerability assessment in foldable displays [14]. Physics-informed neural networks (PINNs) have been applied to programmable origami mechanics [15]. Our work extends this direction by grounding descriptors in neutral-axis mechanics and validating them on physically fabricated stacks with standardized metrology.

2.4 | Agentic AI for Scientific Discovery

Recent advances in LLM agents have shifted scientific computing from passive assistance to active, tool-using systems. Autonomous laboratory agents such as Coscientist [16] and self-evolving agent frameworks [17] demonstrated end-to-end goal decomposition; ChemCrow [18] showed that coupling LLM reasoning with domain tools reduces hallucination. In mechanics, MechAgents [19] demonstrated multi-agent collaborations for simulation. However, existing agentic AI studies typically (i) target generic benchmark problems rather than highly specialized industrial multilayer stacks, (ii) rely predominantly on simulated or synthetic data, (iii) do not connect metrology-defined reliability metrics to interpretable stack descriptors, and (iv) provide limited guidance on blending expert intuition with physics constraints under small-data, proprietary conditions. The present work addresses these gaps.

3 | Methodology

3.1 | Agent-Analytics Workflow

We adopt a constrained human–LLM workflow in which a domain expert defines the engineering hypotheses and the LLM agent serves as a knowledge-structuring mediator to formalize them into mechanics-consistent definitions and to draft descriptor text under confidentiality constraints. Evidence is then quantified with robust ensemble ML, explanation-stability analysis, and pathway analysis under progressive adjustment. In the present extended study, the agent-mediated hypothesis generation was expanded to produce 14 candidate descriptors across five physical mechanism groups. These descriptors were then subjected to a three-stage evidence hierarchy (Section 3.5) to distinguish independently associated drivers from pathway-mediated or insufficient contributors. Specifically, the agent (based on commercial large language models including Claude and Gemini) performed three distinct orchestration functions beyond passive text generation: (i) formalizing qualitative engineering hypotheses into explicit mathematical definitions by systematically applying neutral-axis theory and PAT aggregation across candidate formulations; (ii) generating descriptor variants through iterative human–agent dialogue—for example, evolving the ratio-based S_{sym} into the absolute moment M_{imb} by proposing and evaluating alternative normalizations; and (iii) helping configure the downstream analysis pipeline, including ensemble cross-validation schemes, SHAP computation across all model–fold combinations, and bootstrap adjustment-pathway testing under multiple adjustment conditions. The actual code-level execution and all numerical outputs were independently verified by the domain expert, and all equations, feature computations, and pipeline configurations were script-level verified.

3.2 | Physics-Informed Descriptor Formulation

3.2.1 | Notation

Symbol	Meaning
i	Layer index ($i = 1, \dots, L$)
t_i	Layer thickness
E_i	Elastic modulus (linearized effective)
R_i	Relaxation ratio ($0 = \text{pure elastic}$, $1 = \text{fully relaxed}$)
z_i	Layer centroid coordinate (from bottom reference)
z_{NA}	Neutral-axis location
D_i	PAT-based bending contribution
$\kappa = 1/R$	Folding curvature

3.2.2 | Foundation Equations

The neutral-axis location z_{NA} is determined by the modulus-weighted centroid:

$$z_{NA} = \frac{\sum (E_i t_i z_i)}{\sum (E_i t_i)} \quad (1)$$

The PAT-based bending contribution of each layer is as follows:

$$D_i = E_i \times \left(\frac{t_i^3}{12} + t_i \times (z_i - z_{NA})^2 \right) \quad (2)$$

where the first term represents intrinsic bending stiffness and the second captures lever-arm amplification via the parallel axis theorem. From $\{D_i\}$, we define $D_{top} = \sum D_i$ for $z_i > z_{NA}$ and $D_{bot} = \sum D_i$ for $z_i \leq z_{NA}$. The resulting descriptor family and its mapping to the five physical mechanism groups are summarized in Figure 2.

3.2.3 | Group I: Geometry and Strain (D1–D3)

D1. Total thickness (T_{total}) (3). Total stack thickness governs the lever-arm distance from the neutral axis to each surface. Larger T_{total} increases surface bending strain under folding.

D2. Maximum surface strain proxy (ϵ_{max}) (4). The maximum distance from the neutral axis to any layer surface. Although correlated with T_{total} , ϵ_{max} additionally captures the effect of asymmetric neutral-axis positioning.

D3. Curvature-weighted NA eccentricity (e_e) (5), where $z_{mid} = \frac{(z_{top} + z_{bot})}{2}$ is the geometric centroid. This dimensionless descriptor quantifies how far the neutral axis deviates from the geometric center, amplified by folding curvature.

Descriptor bridge within Group I. The three Group I descriptors are linked by a mechanics-derived bridge equation. From

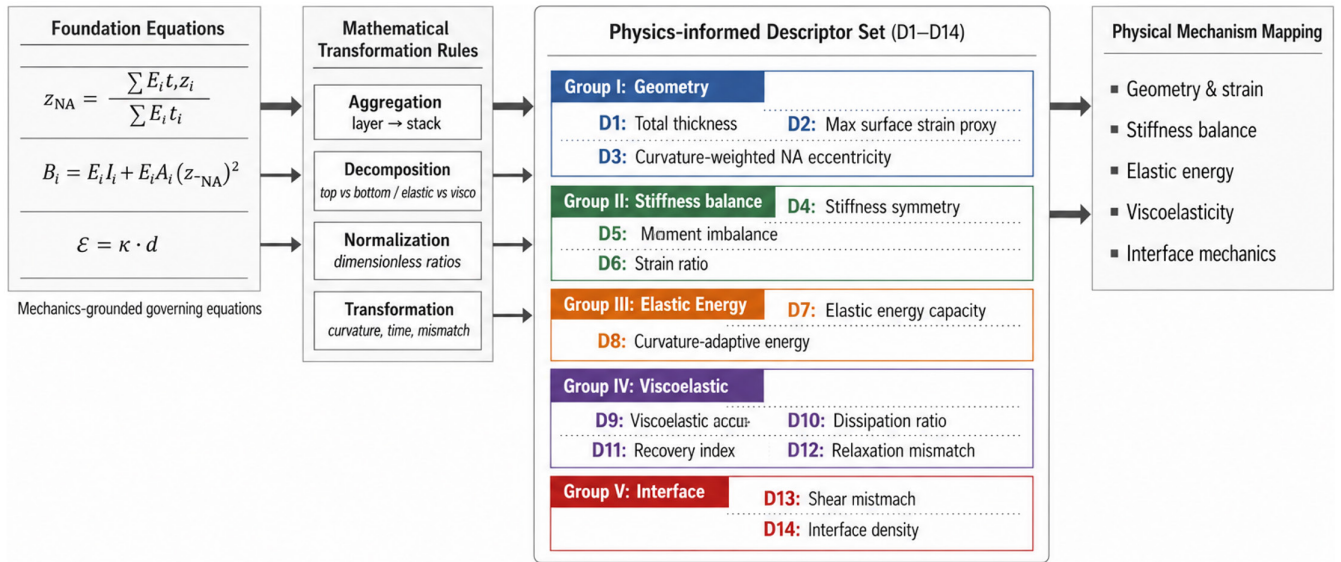


FIGURE 2 | Physics-informed descriptor family: five mechanism groups (D1–D14) derived from foundation equations.

beam geometry, the maximum surface distance from the neutral axis is $d_{\max} = T_{\text{total}}/2 + |z_{NA} - z_{\text{mid}}|$. Under the fixed folding curvature κ used in all experiments, surface strain is proportional to this distance ($\varepsilon_{\max} = \kappa \cdot d_{\max}$), so a linear combination of T_{total} and $|e_{\kappa}|$ should explain the surface strain proxy. Empirically, the fitted bridge $\varepsilon_{\max} = 0.56 \cdot T_{\text{total}} + 591 |e_{\kappa}| - 25.1$ achieves $R^2 = 0.994$ (LOO CV $R^2 = 0.993$, permutation $p < 0.0001$), with the fitted T_{total} coefficient (0.56) closely matching the beam-theory prediction (0.50). The large $|e_{\kappa}|$ coefficient (591) is consistent with the fixed-curvature scaling term $1/\kappa$ because $e_{\kappa} = \kappa(z_{NA} - z_{\text{mid}})$, so $|z_{NA} - z_{\text{mid}}| = |e_{\kappa}|/\kappa$ under the standardized test protocol. Individually, T_{total} and $|e_{\kappa}|$ explain only 30.0% and 11.9% of $\varepsilon_{\max}$ variance, respectively, yet their joint model resolves 99.4%—a suppressor-complementary structure arising from their negative correlation ($r = -0.59$): thicker stacks tend to have more centered neutral axes, creating opposing univariate effects that cancel individually but jointly resolve $\varepsilon_{\max}$ almost exactly (Figures 3 and 4). This confirms that Group I descriptors form a mechanically coherent triad governed by neutral-axis geometry rather than redundant features.

3.2.4 | Group II: Stiffness Balance (D4–D6)

D4. Stiffness symmetry index (S_{sym}) (6). A ratio-based measure of bending stiffness balance about the neutral axis. $S_{\text{sym}} \rightarrow 1$ indicates perfect symmetry.

D5. Dynamic stiffness imbalance (M_{imb}) (7). An advanced descriptor evolved from S_{sym} through continued agent-mediated hypothesis refinement. Whereas S_{sym} captures the ratio, M_{imb} quantifies the absolute moment imbalance normalized by total stiffness—a quantity more directly related to the residual bending moment after unfolding.

To illustrate the human–agent interaction pattern that produced this descriptor refinement, Box 1 presents a representative reconstruction of one such dialogue.

BOX 1 | Representative reconstruction of the human–agent refinement process from S_{sym} to M_{imb} .

Human (framing the physical question):

“The current stiffness-balance descriptor, $S_{\text{sym}} = \min(D_{\text{top}}, D_{\text{bot}})/\max(D_{\text{top}}, D_{\text{bot}})$, is a bounded ratio. Physically, residual crease after unfolding may be more directly related to the bending imbalance that remains around the neutral axis. A ratio-based descriptor may not distinguish a small-absolute-imbalance stack from a large-absolute-imbalance stack when their relative top/bottom stiffness ratios are similar. Can we formulate a descriptor that more directly reflects the absolute stiffness-balance imbalance while remaining comparable across stack configurations?”

Agent (generating candidate normalizations with mechanistic rationale):

Several formulations are possible:

- $|D_{\text{top}} - D_{\text{bot}}|$ —raw absolute imbalance; directly represents the magnitude of top/bottom imbalance but is scale-dependent and therefore less comparable across stacks with different overall stiffness.
- $|D_{\text{top}} - D_{\text{bot}}|/(D_{\text{top}} + D_{\text{bot}})$ —dimensionless and normalized by the top/bottom PAT contributions, but less directly tied to the total stack-level bending contribution.
- $|D_{\text{top}} - D_{\text{bot}}|/\Sigma D_i$ —dimensionless and normalized by total PAT-based stack stiffness; preserves the absolute imbalance signal while improving comparability across configurations.
- $(D_{\text{top}} - D_{\text{bot}})^2/(\Sigma D_i)^2$ —a squared variant that penalizes larger imbalance nonlinearly but loses the linear relationship to imbalance magnitude.

Option (c) is physically motivated because the residual bending response is expected to depend on both the

magnitude of the top-bottom imbalance and the overall stack stiffness scale. Normalization by ΣD_i therefore provides a stack-level imbalance descriptor that remains comparable across configurations.

Human (selecting on physical grounds and specifying verification):

“Option (c) aligns with the interpretation of stiffness-balance imbalance as a primary mechanical driver. Proceed with $M_{imb} = |D_{top} - D_{bot}| / \Sigma D_i$. Verify dimensional consistency, recompute the descriptor across all variants, and compare downstream SHAP ranking and progressive-adjustment behavior against the original S_{sym} formulation.”

† Representative reconstruction of the interaction pattern. Exact prompts were abstracted to omit project-specific material and configuration details; the logical flow and reasoning structure are preserved. All mathematical formulations shown are based on the descriptor definitions introduced in Section 3.2.

This pattern—human-framed physical question, agent-generated mathematical candidates with mechanistic rationale, human selection by physical interpretability, and closed-loop empirical verification—exemplifies the symbolic feature engineering that characterizes the present framework.

D6. Strain extremum ratio (ϵ_{ratio}) (8). The ratio of surface bending strain proxies between the top and bottom outer layers.

(B) Geometry → Strain Bridge: $\epsilon_{max} = a \cdot T_{total} + b \cdot |e_k| + c$

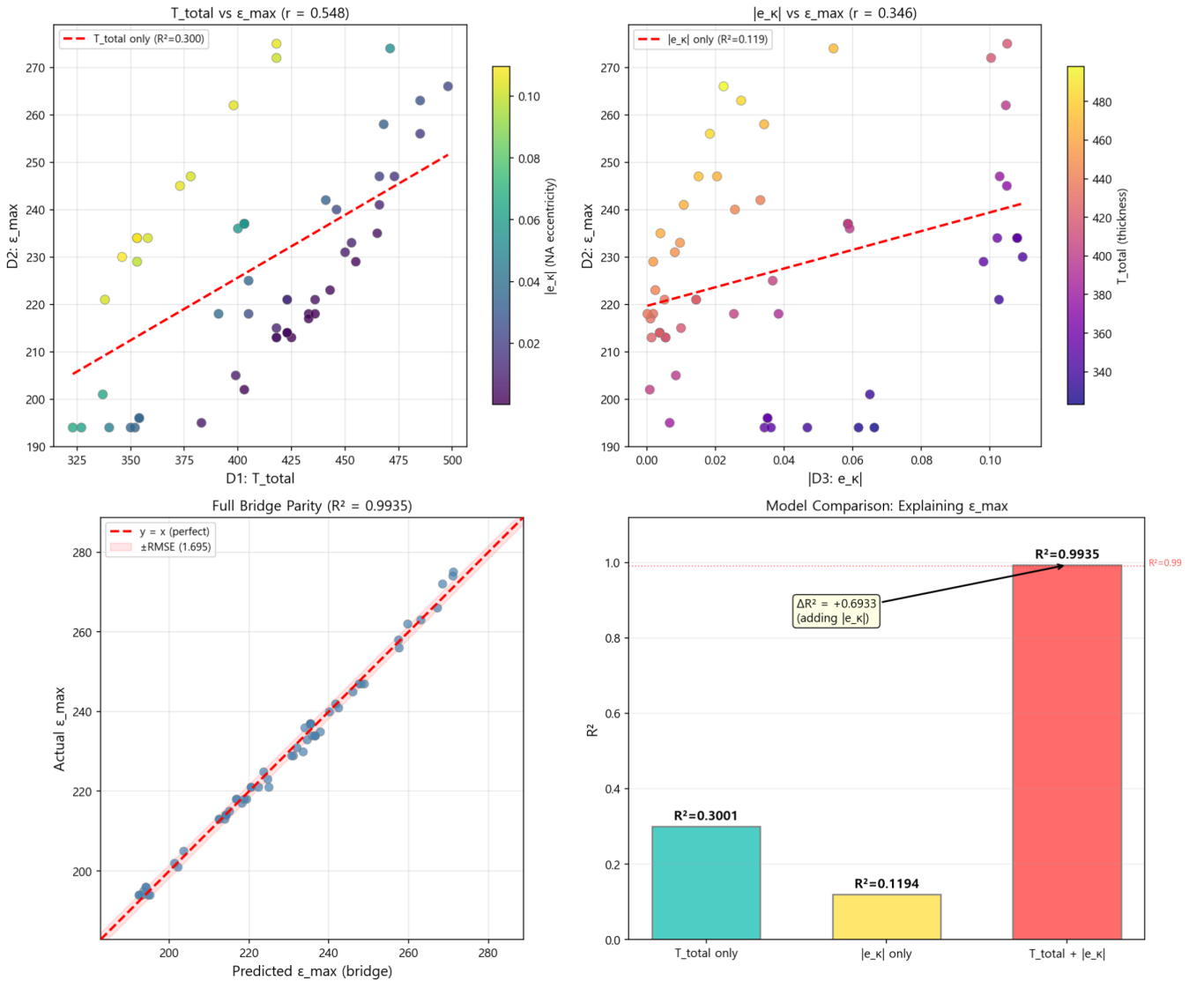


FIGURE 3 | Geometry → strain bridge analysis. (a) T_{total} versus ϵ_{max} colored by $|e_k|$. (b) $|e_k|$ versus ϵ_{max} colored by T_{total} . (c) Bridge parity plot ($R^2 = 0.994$). (d) R^2 comparison showing suppressor-complementary structure (individual $R^2 = 0.30$ and 0.12 , joint $R^2 = 0.994$).

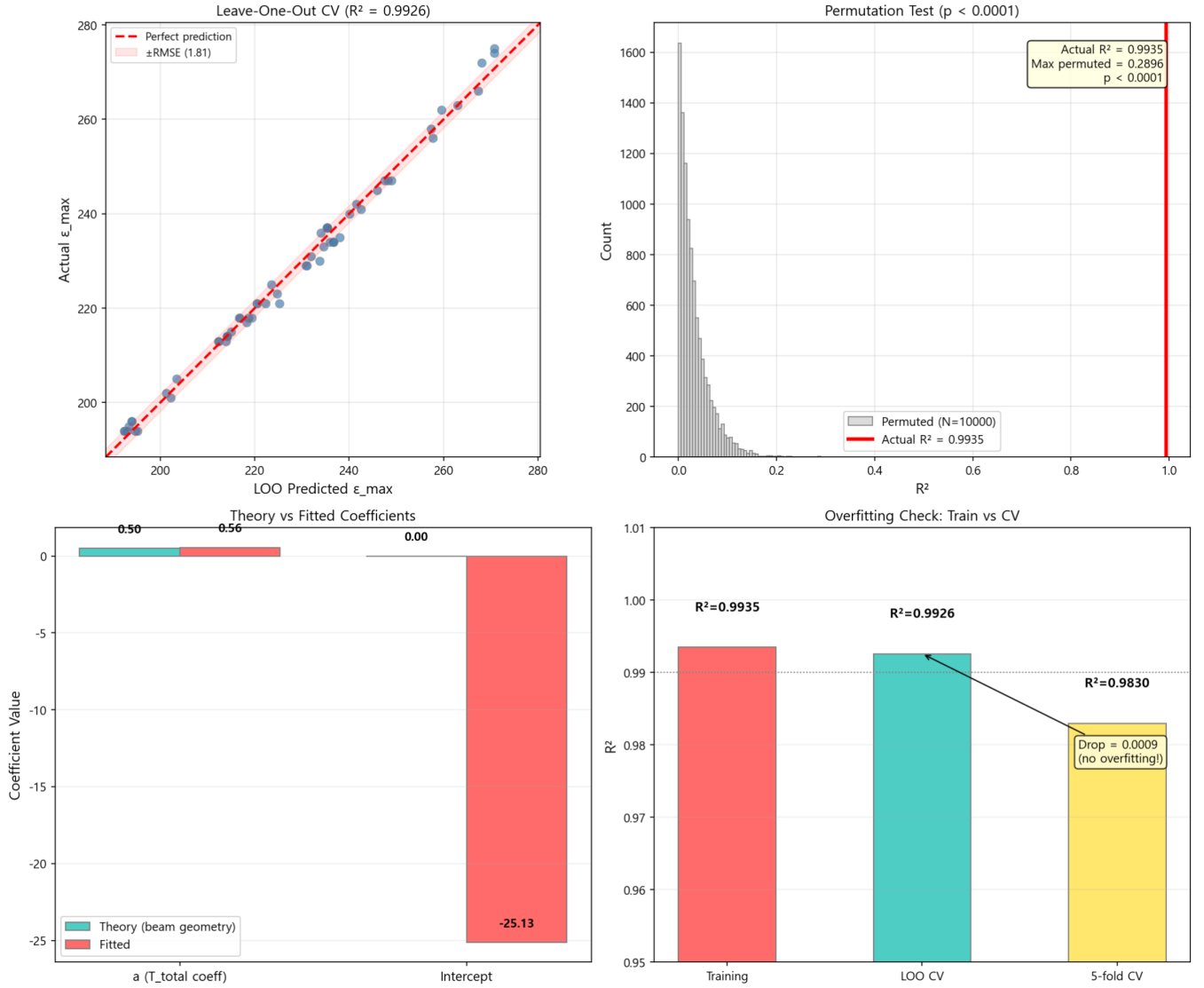


FIGURE 4 | Bridge equation verification. (a) Leave-one-out CV parity ($R^2=0.993$). (b) Permutation test ($p < 0.0001$, $N = 10,000$). (c) Theory versus fitted coefficients ($a_{theory}=0.50$, $a_{fit}=0.56$). (d) Overfitting check (Training/LOO/5-fold R^2 comparison, drop = 0.0009).

3.2.5 | Group III: Elastic Energy (D7–D8)

D7. Elastic strain-energy capacity (U_{el}) (9). The total bending stiffness contribution from predominantly elastic layers ($E_i > E_{thresh}$ and $R_i < R_{thresh}$). Higher U_{el} implies greater restoring capacity upon unfolding.

D8. Curvature-adaptive energy (U_{ca}) (10). An evolution of U_{el} that incorporates folding curvature. Under constant folding radius, U_{ca} is linearly proportional to U_{el} .

3.2.6 | Group IV: Viscoelastic (D9–D12)

D9. Viscoelastic accumulation (V_{ve}) (11). The central viscoelastic descriptor, combining creep compliance proxy ($\frac{t}{E}$) with relaxation severity (R). Higher V_{ve} indicates greater time-dependent deformation accumulation.

D10. Viscoelastic energy dissipation ratio (η_{diss}) (12). A dimensionless ratio (0–1) quantifying the fraction of total bending stiffness subject to relaxation.

D11. Nonlinear recovery index (R_{nl}) (13). Incorporates time-dependent logarithmic decay of recovery capacity. Here, $\alpha = 0.1$ is the logarithmic decay coefficient, and $t_{hold} = 60s$ is the folding hold duration, both fixed across all variants to match the standardized test protocol.

D12. Relaxation mismatch index (Λ_{relax}) (14). Captures the heterogeneity of relaxation across adjacent interfaces.

3.2.7 | Group V: Interface Mechanics (D13–D14)

D13. Interlayer shear mismatch index (τ_{mis}) (15). A shear-oriented descriptor complementing the bending-dominated Groups I–IV.

D14. Interface density parameter (ρ_{int}). (16). The product of interface density and mean modulus contrast across interfaces.

To preserve industrial confidentiality, absolute stack compositions are not disclosed; all modeling and reporting rely on derived descriptors, standardized effect sizes, and robustness outcomes.

3.3 | Ensemble Modeling and Explanation Stability

We train an ensemble of 14 heterogeneous regression models (random forest, gradient boosting, extra trees, AdaBoost, Ridge, Lasso, ElasticNet, SVR-RBF, SVR-Linear, KNN, Gaussian process, CatBoost, LightGBM, and XGBoost) to reduce model-specific bias under small-data conditions. All preprocessing is fit on training folds only, and performance is reported with fivefold cross-validation. Global explanations are computed using SHAP and aggregated across models and folds. Attribution uncertainty is quantified by bootstrap confidence intervals ($B=50$; the relatively modest B is sufficient because the ensemble-fold aggregation across 14 models $\times$ 5 folds already reduces attribution variance substantially).

3.4 | Crease Metrology: Crease Max Slope

Crease severity is quantified by crease max slope ($\mu\text{m}/\text{mm}$), defined as the maximum magnitude of the first derivative of a crease-profile cross section. A 2D height map is measured over a fixed ROI centered at the crease apex. For each design variant, the modeling target is the median across repeated measurements. The measurement definition and profile-based extraction procedure are illustrated in Figure 5.

3.5 | Three-Stage Evidence Hierarchy

To distinguish predictive association from independent contribution within the proposed adjustment hierarchy, we replace single-framework causal-consistency analysis with a progressive three-stage evidence hierarchy. In the present observational dataset, this analysis should be interpreted as adjustment-based causal consistency rather than definitive interventional causality.

3.5.1 | Stage 1: Ensemble SHAP With Bootstrap Confidence Intervals

Global feature importance is computed via SHAP across all ensemble members and cross-validation folds. Importance stability is assessed through bootstrap resampling ($B=50$) to produce 95% confidence intervals. This stage ranks all 14 descriptors by predictive contribution.

3.5.2 | Stage 2: Pathway Analysis Under Progressive Adjustment via Bootstrap ATE

For each descriptor, we estimate the average treatment effect (ATE) on crease max slope under five progressive adjustment conditions: (a) bivariate, (b) adjusted for geometry (T_{total}), (c) adjusted for viscoelasticity (V_{ve}), (d) adjusted for stiffness balance (M_{imb}), and (e) full adjustment (all three). Each condition uses bootstrap resampling ($B=1000$) with 95% confidence intervals. Three orthogonal base mechanisms—total thickness (geometric pathway), viscoelastic accumulation (material pathway), and stiffness imbalance (structural balance pathway)—constitute the minimal sufficient adjustment set, ensuring adequate degrees of freedom ($N=63$, $df \geq 59$). The selection of these three variables as the minimal sufficient adjustment set is justified on both physical and empirical grounds. Physically, they represent three orthogonal crease-formation mechanisms: geometric lever-arm amplification (thickness), time-dependent irreversible deformation (viscoelasticity), and residual bending-moment balance (stiffness imbalance pathway). Empirically, their pairwise correlations are low ($|r| < 0.31$; Figure 7), and they emerge as the top-ranked descriptors from three independent mechanism groups in the SHAP analysis (Groups I, IV, and II, respectively).

3.5.3 | Stage 3: Evidence-Tier Classification

Based on Stages 1 and 2, each descriptor is assigned to one of four evidence tiers. Two classification override rules refine this scheme. Override 1: When near-equivalent descriptors ($|r| > 0.99$) both reach full-adjustment significance, only the higher-SHAP-ranked representative is promoted to Tier 1; the other is classified as Tier 3 (superseded). Override 2: Descriptors that become significant only after adjustment, without bivariate significance, are conservatively retained as Tier 4 (suppression-pattern candidates) pending validation in larger datasets.

Tier	Criteria	Interpretation
Tier 1	SHAP Top 5 + full adj. significant	Independently associated driver
Tier 2	SHAP Top 5 + bivariate only	Pathway-mediated contributor
Tier 3	Bivariate significant (any rank)	Associated, potentially superseded
Tier 4	No robust bivariate support or adjustment-only significance (suppressor candidate)	Insufficient evidence

3.6 | Uncertainty Quantification

Prediction uncertainty is quantified at two levels: (i) global model disagreement, measured as the standard deviation across ensemble members (mean RSD=6.9%), and (ii) local

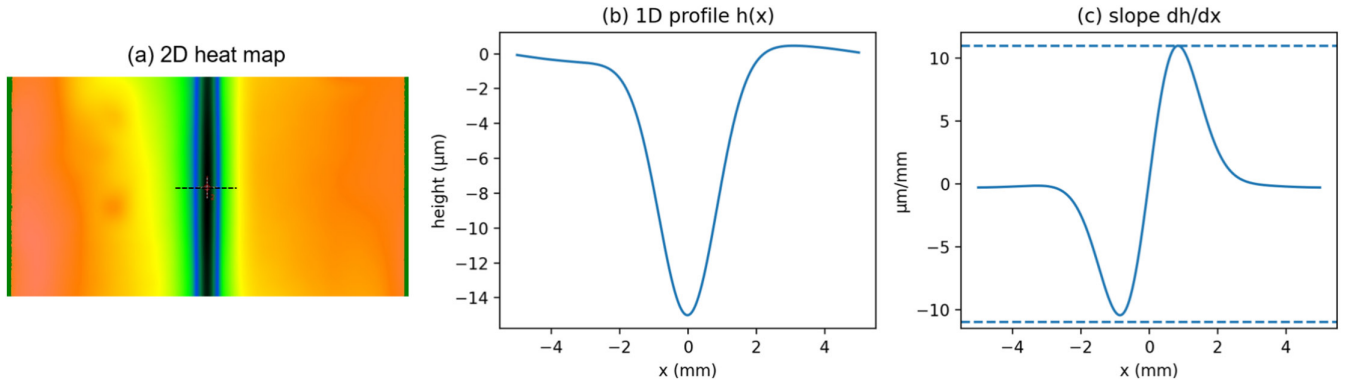


FIGURE 5 | Definition of crease max slope ($\mu\text{m}/\text{mm}$) from 2D surface profilometry and a cross-sectional profile $h(x)$.

inconsistency, estimated via KNN leave-one-out error bars ($k=3$). The consistently low RSD (98.4% of samples below 20%) indicates robust model consensus across the descriptor space.

4 | Results and Discussion

Interpretation note: Results are evaluated by physics consistency and by statistical/causal analysis, not by narrative generation.

4.1 | Dataset Summary

Under the wing-support condition (industry-aligned metrology), the observed distribution of crease max slope shows a median of $12.561 \mu\text{m}/\text{mm}$ and a mean of $13.076 \mu\text{m}/\text{mm}$, with a best (minimum) value of $4.480 \mu\text{m}/\text{mm}$ and worst of $28.830 \mu\text{m}/\text{mm}$ (Table 1). Each data point represents the per-variant median crease max slope, ensuring statistical independence. Importantly, each of the 63 data points represents a physically distinct multilayer stack—fabricated with different material combinations and layer thicknesses—not repeated measurements of a single design. To mitigate the inherent measurement variability of crease profilometry, the modeling target for each variant is the median across multiple repeated measurements, ensuring that individual metrology noise does not dominate the design-level signal.

4.2 | Quartile-Stratified Validation

Quartile-stratified analysis was performed at the design-variant level ($N=63$). Each data point represents an independent design condition, avoiding pseudo-replication from repeated measurements. For the three Tier 1 descriptors plus the Tier 2 descriptor, samples were grouped into quartiles (Q1–Q4), and crease max slope distributions were compared using the Kruskal–Wallis test (Figure 6).

Viscoelastic accumulation (D9) exhibits the strongest monotonic trend: Q1 $\rightarrow$ Q4 mean crease increases by 77.1% ($p = 4.6 \times 10^{-9}$). Total thickness (D1) shows a 37.5% increase ($p = 2.6 \times 10^{-5}$). Dynamic stiffness imbalance (D5) shows a 27.7% increase approaching but not reaching conventional significance in univariate stratification ($p=0.080$). This is

TABLE 1 | Summary statistics of crease max slope ($\mu\text{m}/\text{mm}$) under the wing-support configuration.

Median	Mean	Min	Max	SD	Unit
12.561	13.076	4.480	28.830	3.861	$\mu\text{m}/\text{mm}$

consistent with the pathway analysis finding that M_{imb} 's marginal effect is partially confounded by S_{sym} ($r = -0.997$) in unadjusted analyses while achieving independent significance under multivariate progressive adjustment (Tier 1). Maximum surface strain proxy (D2) reaches significance at $p=0.044$ with a 23.0% increase.

4.3 | Multicollinearity Diagnostic

Prior to modeling, we assessed descriptor interdependence (Figure 7). Two notable relationships were observed. First, D4 (S_{sym}) and D5 (M_{imb}) exhibit near-perfect anticorrelation ($r = -0.997$), confirming their mathematical relationship as complementary representations of stiffness balance. Second, D7 (U_{el}) and D8 (U_{ca}) are perfectly correlated ($r=1.000$), which is expected since $U_{ca} = \frac{1}{2}U_{el}\kappa^2$ and the folding radius is constant across all experiments. A cluster of high correlations ($r=0.85$ – 0.95) among D11, D12, and D13 reflects the physical coupling between relaxation-weighted stiffness, relaxation heterogeneity, and shear mismatch. Despite elevated VIF values for several descriptors, all 14 were retained for the SHAP ranking stage; the pathway analysis under progressive adjustment subsequently provides a principled mechanism for separating independent from redundant contributions.

4.4 | Ensemble Model Performance

The heterogeneous ensemble of up to 14 regression models was evaluated under fivefold cross-validation. CatBoost achieved the highest mean R^2 (0.710), followed by SVR-RBF (0.707). The cross-model agreement (12 of 14 models achieving $R^2 > 0.59$) indicates robust predictive signal. A parsimonious model using only the four Tier 1 + 2 descriptors retained 91% of the full-model performance ($R^2=0.644$), confirming that the three base mechanisms capture the dominant predictive

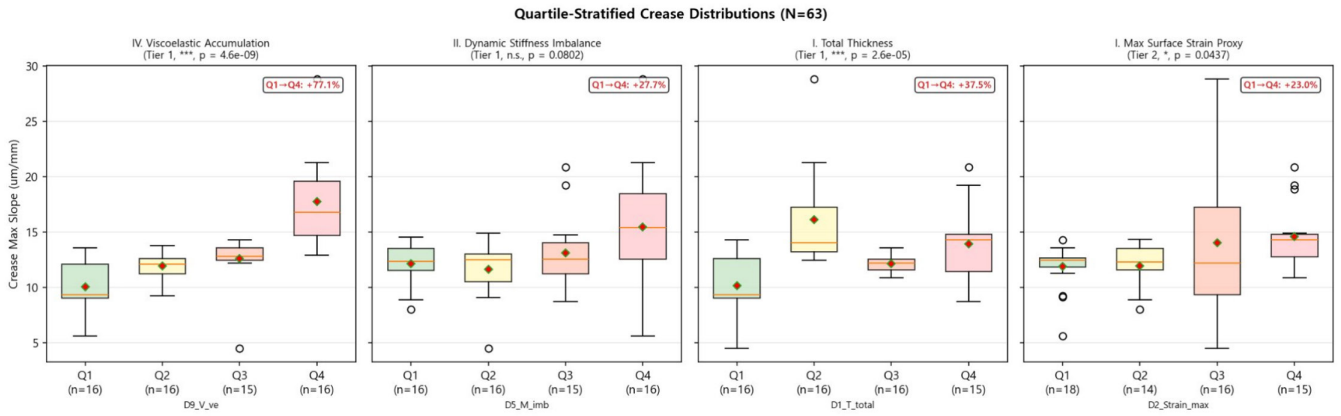


FIGURE 6 | Quartile-stratified crease distributions ($N=63$) for Tier 1 and Tier 2 descriptors.

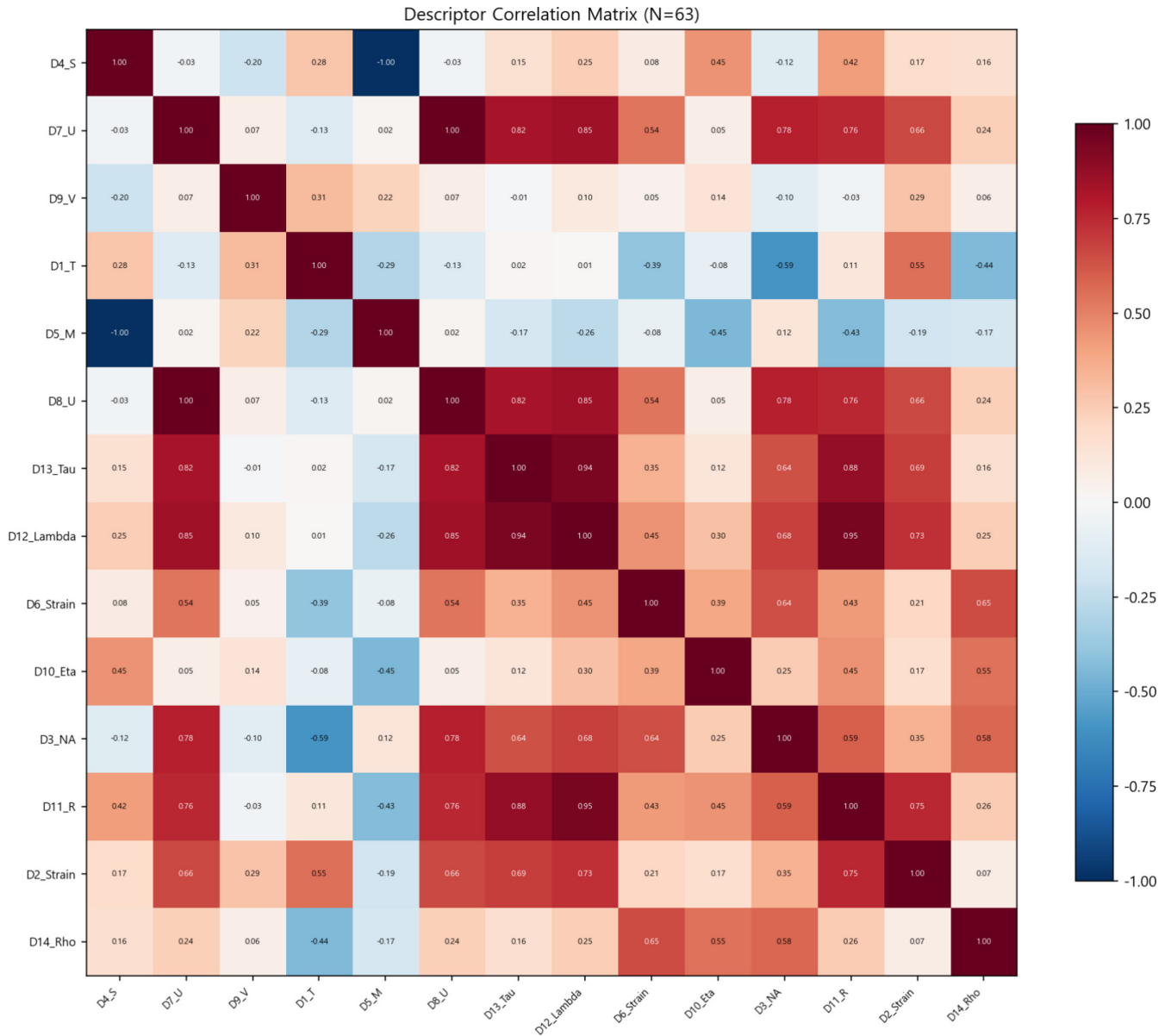


FIGURE 7 | Descriptor correlation matrix ($N=63$).

signal. The parsimonious four-feature model (Tier 1 + 2 only) retaining 91% of full-model performance ($R^2=0.644$ vs. 0.710) argues against a purely noise-driven interpretation

of the descriptor model: If the full model were primarily fitting noise rather than capturing physical signal, performance would degrade sharply upon feature reduction from 14 to 4.

We additionally note that a black-box baseline using all 61 raw layer-level properties achieves $R^2=0.87$; however, this operates at a feature-to-sample ratio of 0.97 (61/63), where tree-based models may exploit high-dimensional sample-specific partitions, making cross-validated performance more optimistic under small- N conditions. More fundamentally, the objective of the present framework is not to maximize predictive accuracy but to identify interpretable, adjustment-robust physical mechanisms that provide actionable design guidance. A 61-feature model cannot answer “why does this stack crease?” or “what should a designer change?”—questions that the three Tier 1 descriptors answer directly. The information compression from 61 raw properties to a low-dimensional physics-informed descriptor space inevitably entails some loss in predictive resolution; however, this trade-off yields mechanistic interpretability, mechanistic traceability, and prescriptive utility that raw-feature models cannot provide.

4.5 | Descriptor Sufficiency and Predictive Gap

The predictive gap between the 14-descriptor ensemble ($R^2 \approx 0.71$) and the raw-feature black-box baseline ($R^2 \approx 0.87$) warrants explicit discussion. We do not interpret the proposed descriptor set as exhaustive. Rather, the 14 descriptors are intended to capture the dominant, interpretable, and adjustment-robust mechanisms under small-data industrial constraints.

First, the raw-feature baseline operates at a feature-to-sample ratio close to unity (61 raw properties over 63 design variants), a regime in which tree-based ensembles may exploit high-dimensional sample-specific partitions, leading to more optimistic cross-validated performance under small- N conditions. In contrast, the descriptor-based model reduces this ratio to 0.22 (14 descriptors over 63 variants), providing a more conservative and mechanism-focused representation. A portion of the apparent R^2 gap may therefore reflect the different statistical regimes of the two models rather than purely additional transferable physical information.

Second, raw layer-level features retain microscale information that aggregate descriptors necessarily compress. The present descriptor formulations group layers into neutral-axis-based top/bottom partitions or sum over the stack, whereas raw features preserve layer-specific identity. This allows black-box models to exploit layer-pair-specific interactions, such as adhesive–substrate coupling at particular stack positions, which are only represented in aggregate form by the present interface descriptors.

Third, layer-ordering effects beyond modulus-weighted centroid aggregation may also contribute. Stacks with similar total thickness and neutral-axis location but different soft/hard layer sequences may exhibit different shear-lag, local stress concentration, or interfacial slip behavior. These effects are

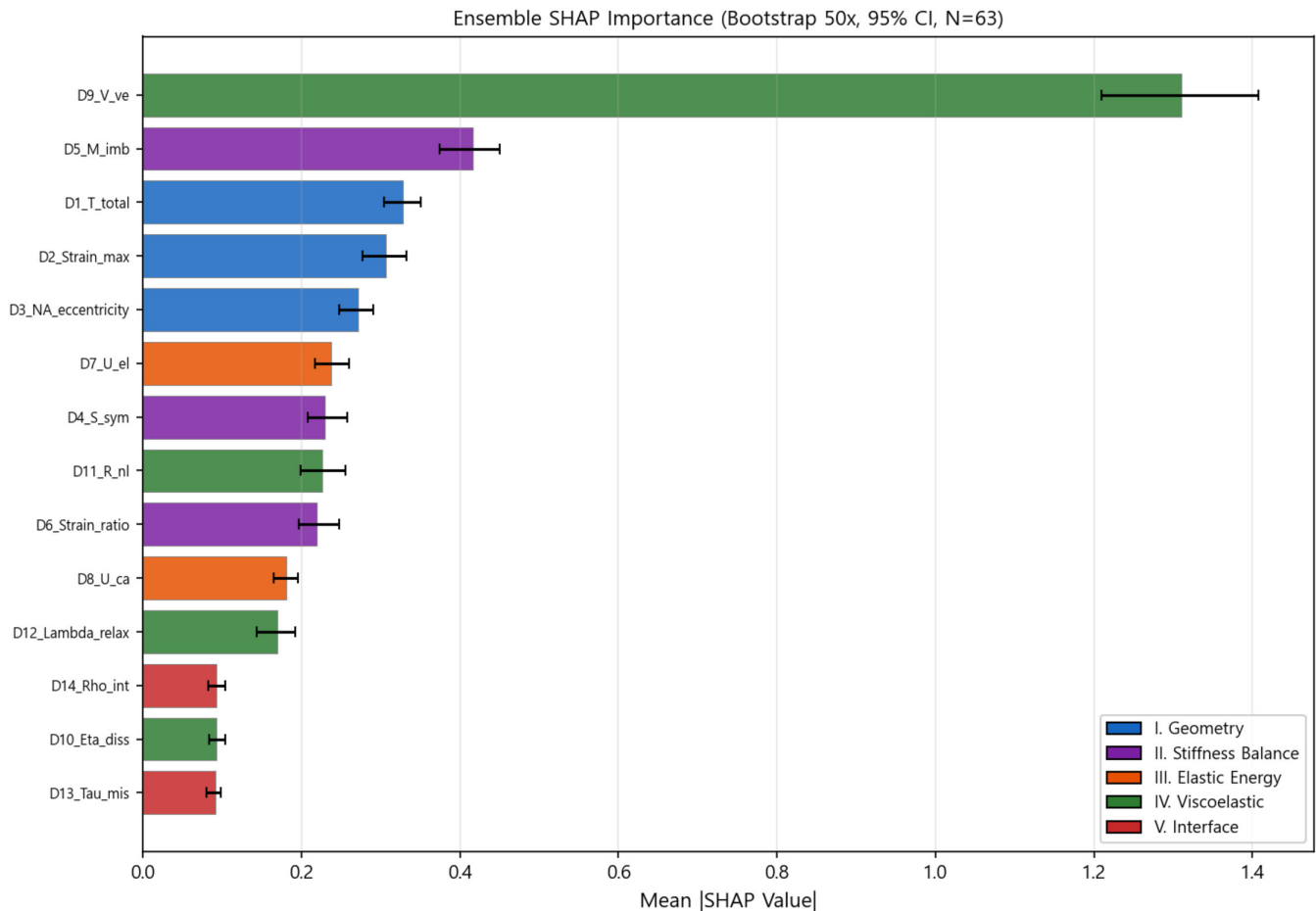


FIGURE 8 | Ensemble SHAP importance ranking with bootstrap 95% CI. Colors indicate mechanism groups.

mechanistically plausible in heterogeneous multilayer stacks but cannot be isolated reliably from the present dataset alone.

We therefore interpret the R^2 difference as a combined effect of high-dimensional statistical flexibility, layer-specific micro-interactions, layer-ordering information, and residual process/metrology variance. The objective of the proposed framework is not to maximize predictive accuracy, but to identify interpretable and design-actionable mechanisms. In this sense, the compression from raw properties to physics-grounded descriptors is an intentional methodological trade-off: Some predictive ceiling is sacrificed in exchange for mechanistic traceability, evidence-tier classification, and prescriptive design guidance.

4.6 | Global Interpretability via SHAP

The ensemble SHAP ranking with bootstrap 95% confidence intervals (Figure 8) reveals a clear hierarchy. Viscoelastic accumulation (D9, V_{ve}) dominates with a mean |SHAP| of 1.293—approximately three times the second-ranked descriptor. Dynamic stiffness imbalance (D5, M_{imb}) ranks second (0.446), followed by total thickness (D1, 0.360), maximum strain proxy (D2, 0.319), and neutral-axis eccentricity (D3, 0.311).

Examining the ranking by mechanism group reveals that Group IV (viscoelastic) contributes the strongest single predictor (D9), Group II (stiffness balance) provides the second

strongest (D5), and Group I (geometry) contributes three of the top five. Group III (elastic energy) and Group V (interface) rank lower, suggesting that their crease-related information is largely captured by the higher-ranked descriptors. Notably, S_{sym} (D4) ranks 9th, substantially below its evolved counterpart M_{imb} (D5, rank 2).

4.7 | Pathway Analysis Under Progressive Adjustment

To distinguish predictive association from independent contribution within the proposed adjustment hierarchy, we performed bootstrap pathway analysis under progressive confounder adjustment ($B = 1000$, 95% CI; Figure 9).

- Tier 1—Independently robust descriptors. The three base mechanism descriptors—D9 (viscoelastic accumulation), D5 (dynamic stiffness imbalance), and D1 (total thickness)—constitute the minimal sufficient adjustment set. They were selected as base-mechanism descriptors on physical and empirical grounds, and each satisfies the Tier 1 criterion under the implemented adjustment scheme (when a descriptor is itself a member of the adjustment set, it is excluded from its own confounder list to avoid self-adjustment). Among them, D9 exhibits the most robust pattern: significant across all adjustment conditions (bivariate: +0.71, Adj_Geometry: +0.67, Adj_Visco: +0.72, Adj_Balance: +0.65). D1 shows

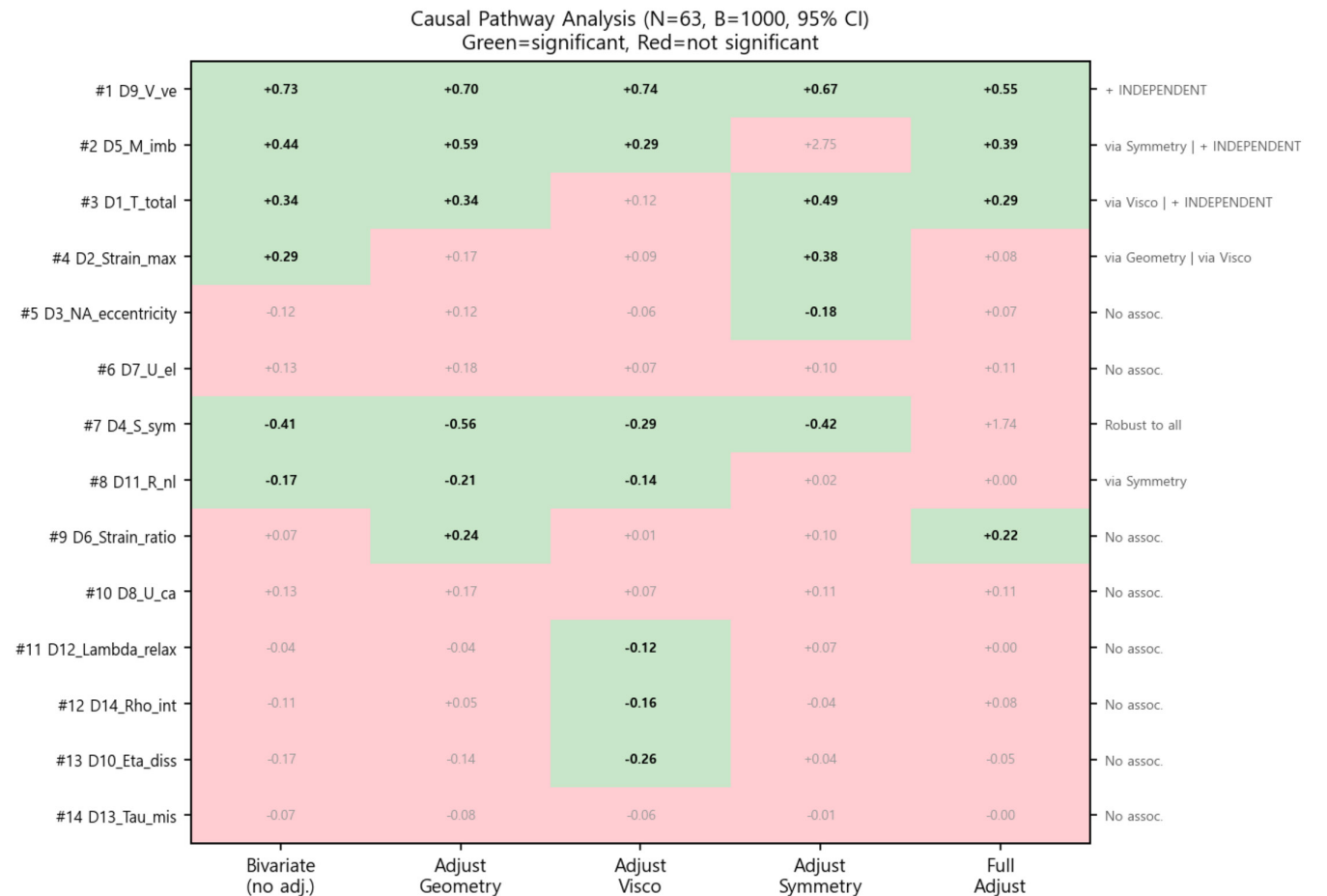


FIGURE 9 | Adjustment-pathway heatmap. Green = significant (95% CI), red = not significant. Right column shows identified pathway.

partial mediation through the viscoelastic pathway (bivariate: +0.34, Adj_Visco: +0.14, nonsignificant), indicating that thickness contributes through both a retained geometric pathway and a mediated viscoelastic pathway.

- Tier 2—Pathway-mediated. D2 (ϵ_{max} , SHAP rank 4) was significant bivariately (ATE=+0.29) but lost significance under both geometry and viscoelastic adjustment, indicating it operates through the T_{total} and V_{ve} pathways.
- Tier 3—Associated (superseded). D4 (S_{sym}) was significant bivariately (ATE=−0.41) but sign-reversed and nonsignificant under full adjustment (+1.74), indicating coefficient instability under collinearity with D5. However, D4 and D5 are near-perfectly anticorrelated ($r = -0.997$), representing complementary encodings of the same physical mechanism (stiffness balance). When near-equivalent descriptors both reach significance under mutual adjustment, coefficient instability (suppressor effects) rather than independent identification is the more parsimonious explanation. We therefore retain D5 (M_{imb}) as the representative Tier 1 descriptor for stiffness balance on the basis of its superior SHAP ranking (#2 vs. #9) and classify D4 as Tier 3 (superseded). D11 (R_{nl}) showed a significant bivariate effect (−0.17) mediated through the balance pathway.
- Tier 4—Insufficient evidence. The remaining descriptors did not reach bivariate significance. D6 (ϵ_{ratio}) exhibited a notable suppressor pattern: nonsignificant bivariately but significant under full adjustment, suggesting that its effect emerges only after partialling out confounders. Although statistically

interesting, we conservatively classify such cases as Tier 4 in the absence of bivariate support, noting them as candidates for investigation in larger datasets. This classification does not imply physical irrelevance; it indicates that these descriptors' crease-related information is largely captured by the three Tier 1 mechanisms at the current sample size ($N=63$).

The bootstrap bivariate ATE estimates and their 95% confidence intervals are summarized in Figure 10, whereas the complete evidence-tier classification for all 14 descriptors is summarized in Table 2.

4.8 | Descriptor Evolution: From S_{sym} to M_{imb}

The evolution from ratio-based stiffness symmetry (D4, S_{sym}) to absolute moment imbalance (D5, M_{imb}) is a central finding of the agent-analytics framework. This hypothesis is now supported by three independent lines of evidence:

- Predictive improvement. Substituting the ratio-based stiffness-symmetry formulation with the absolute moment-imbalance formulation in the ensemble yields a +1.75% improvement in cross-validated R^2 while simultaneously reducing RMSE and MAE.
- SHAPrankingshift. M_{imb} ranks 2nd (mean |SHAP| = 0.446), whereas S_{sym} ranks 9th (0.210), indicating that the ensemble extracts substantially more predictive information from the absolute formulation.

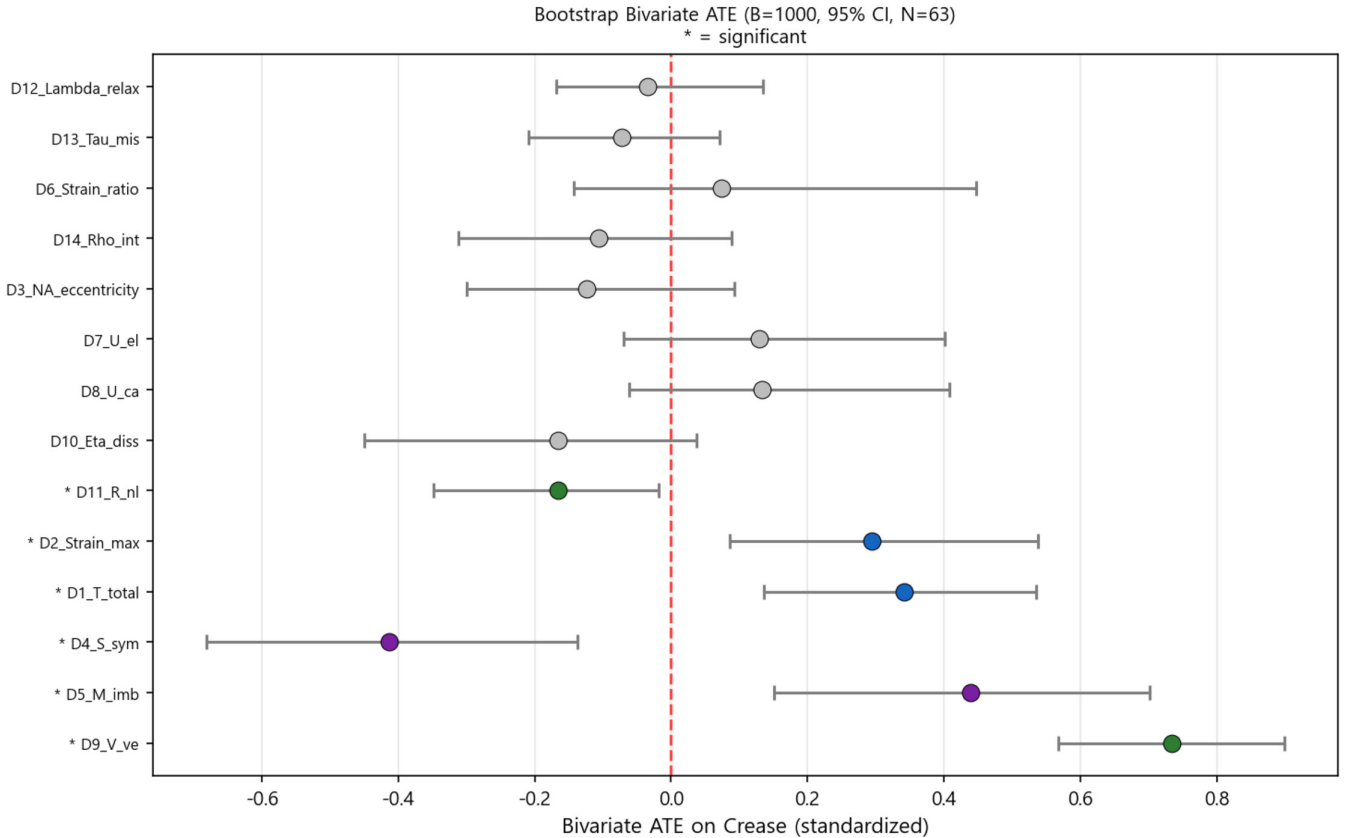


FIGURE 10 | Bootstrap bivariate ATE forest plot (B=1000, 95% CI, N=63).

TABLE 2 | Evidence summary.

Rank	Descriptor	Group	Biv ATE	Full Sig	Evidence tier
1	D9 (V_{ve})	IV. Viscoelastic	+0.710	Yes	Tier 1: Independently associated
2	D5 (M_{imb})	II. Balance	+0.432	Yes	Tier 1: Independently associated
3	D1 (T_{total})	I. Geometry	+0.355	Yes	Tier 1: Independently associated
4	D2 (ϵ_{max})	I. Geometry	+0.284	No	Tier 2: Pathway-mediated
5	D3 (e_k)	I. Geometry	-0.149	No	Tier 4: Insufficient
6	D7 (U_{el})	III. Elastic	+0.110	No	Tier 4: Insufficient
7	D6 (ϵ_{ratio})	II. Balance	+0.058	Yes	Tier 4: Insufficient
8	D12 (Λ_{relax})	IV. Viscoelastic	-0.058	No	Tier 4: Insufficient
9	D4 (S_{sym})	II. Balance	-0.405	No	Tier 3: Associated
10	D8 (U_{ca})	III. Elastic	+0.114	No	Tier 4: Insufficient
11	D11 (R_{nl})	IV. Viscoelastic	-0.171	No	Tier 3: Associated
12	D13 (τ_{mis})	V. Interface	-0.096	No	Tier 4: Insufficient
13	D14 (ρ_{int})	V. Interface	-0.134	No	Tier 4: Insufficient
14	D10 (η_{diss})	IV. Viscoelastic	-0.162	No	Tier 4: Insufficient

iii. Evidence-tier elevation. Under the three-stage evidence hierarchy, D5 (dynamic stiffness imbalance, M_{imb}) achieves Tier 1 (independently associated under full adjustment), whereas D4 (stiffness symmetry index, S_{sym}) is classified as Tier 3 (associated but superseded). This qualitative shift in evidence-tier status provides convergent evidence from three methods: D5 provides a more stable and practically informative parameterization under the present dataset and evidence hierarchy—the absolute residual bending moment—that D4’s ratio-based formulation does not equivalently capture.

The perfect rank-order anti-correlation (Spearman $\rho = -1.000$, Pearson $r = -0.997$) and the near-exact polynomial fit ($R^2 = 1.000$ under quadratic regression) confirm that S_{sym} and M_{imb} are strict monotone-transform equivalents, encoding the same stiffness-balance mechanism with different normalizations: S_{sym} as a ratio and M_{imb} as a total-stiffness-normalized absolute difference. The near-perfect anti-correlation ($r = -0.997$) between D4 and D5 confirms their mathematical kinship. However, the divergence in evidence-tier status demonstrates that even small reformulations of physics descriptors can yield qualitatively different insights within the proposed evidence hierarchy, supporting the interpretation that M_{imb} is a more informative descriptor than S_{sym} for the present dataset and adjustment framework.

4.9 | Case Study: Geometric Compensation

We analyzed three representative design variants using ensemble SHAP waterfall plots and a SHAP-weighted PCA design-space projection (Figure 11) to illustrate the prescriptive utility of the framework.

- Worst case (#4, crease = 20.9 $\mu\text{m}/\text{mm}$). All three Tier 1 drivers are aligned in the crease-promoting direction: V_{ve} contributes +2.8, T_{total} +1.5, and M_{imb} +0.7 to the predicted crease. This represents a thick, viscoelastically heavy, and structurally unbalanced stack.
- Extreme optimization (#18, crease = 5.6 $\mu\text{m}/\text{mm}$). V_{ve} (-2.5) and T_{total} (-1.8) are aggressively minimized. However, M_{imb} (+1.0) reveals severe stiffness imbalance. Although this configuration achieves low crease, it violated non-crease mechanical reliability requirements, making it unsuitable under the broader mechanical reliability requirements used in development screening.
- Balanced design (#56, crease = 8.0 $\mu\text{m}/\text{mm}$). All three Tier 1 descriptors plus D2 (Tier 2) contribute moderately in the crease-reducing direction: V_{ve} (-1.5), T_{total} (-0.8), M_{imb} (-0.3), and D2 (-0.2). Crucially, no single descriptor is pushed to extremes—unlike #18, where V_{ve} and T_{total} are aggressively minimized at the cost of structural balance. The key differentiator is M_{imb} : #56 achieves -0.3 (balanced) versus #18’s +0.9 (severely unbalanced). Although #56’s absolute crease (8.0) is higher than #18’s (5.6), it represents a practically deployable design that better balances crease performance with broader structural reliability considerations. This demonstrates that the “balanced design” strategy trades a modest crease penalty for holistic design viability—a trade-off that is essential in industrial foldable display development. The SHAP-weighted PCA projection (Figure 11, right panel) provides spatial confirmation: #4 occupies the high-crease region (right, along the V_{ve} loading direction), #18 is isolated in the extreme low- V_{ve} zone (far left), and #56 resides within the low-crease contour band alongside neighboring designs with comparable crease levels, confirming that its performance is consistent with its descriptor-space neighborhood rather than an outlier artifact.

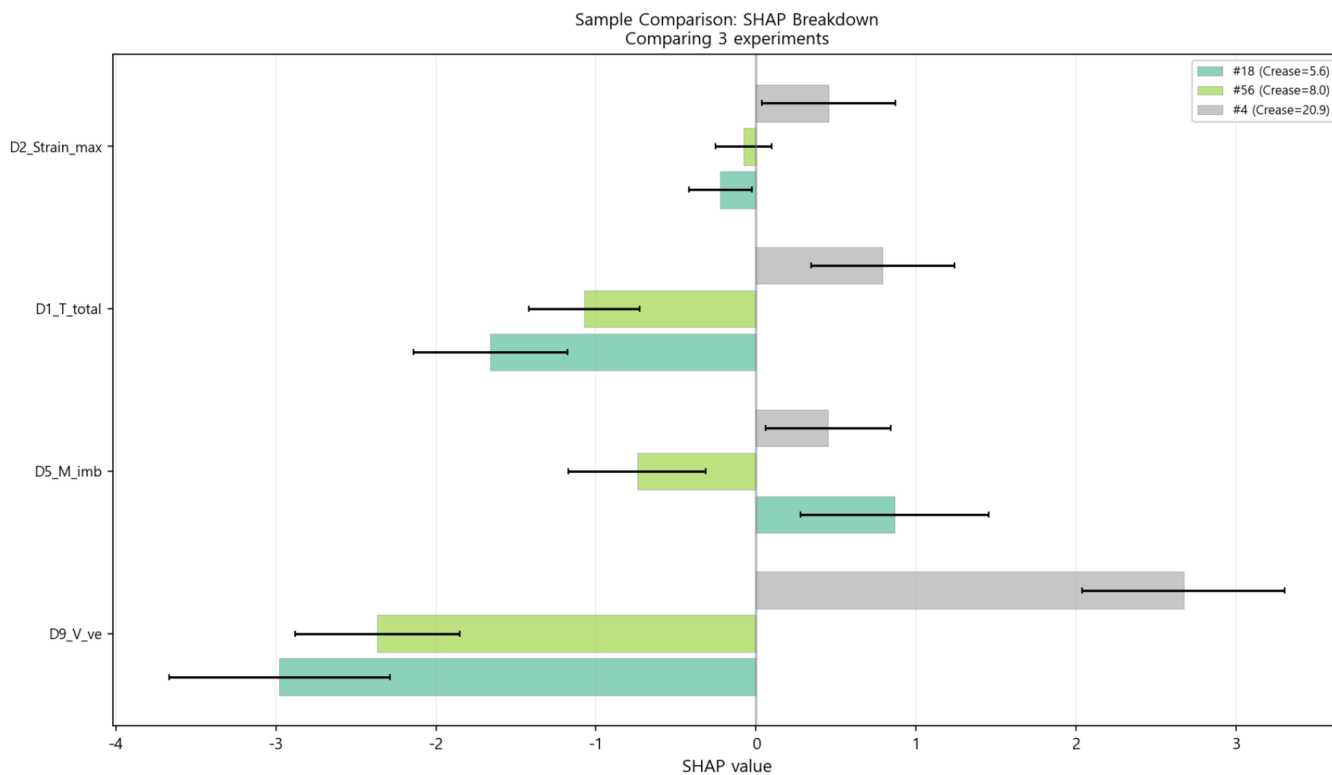
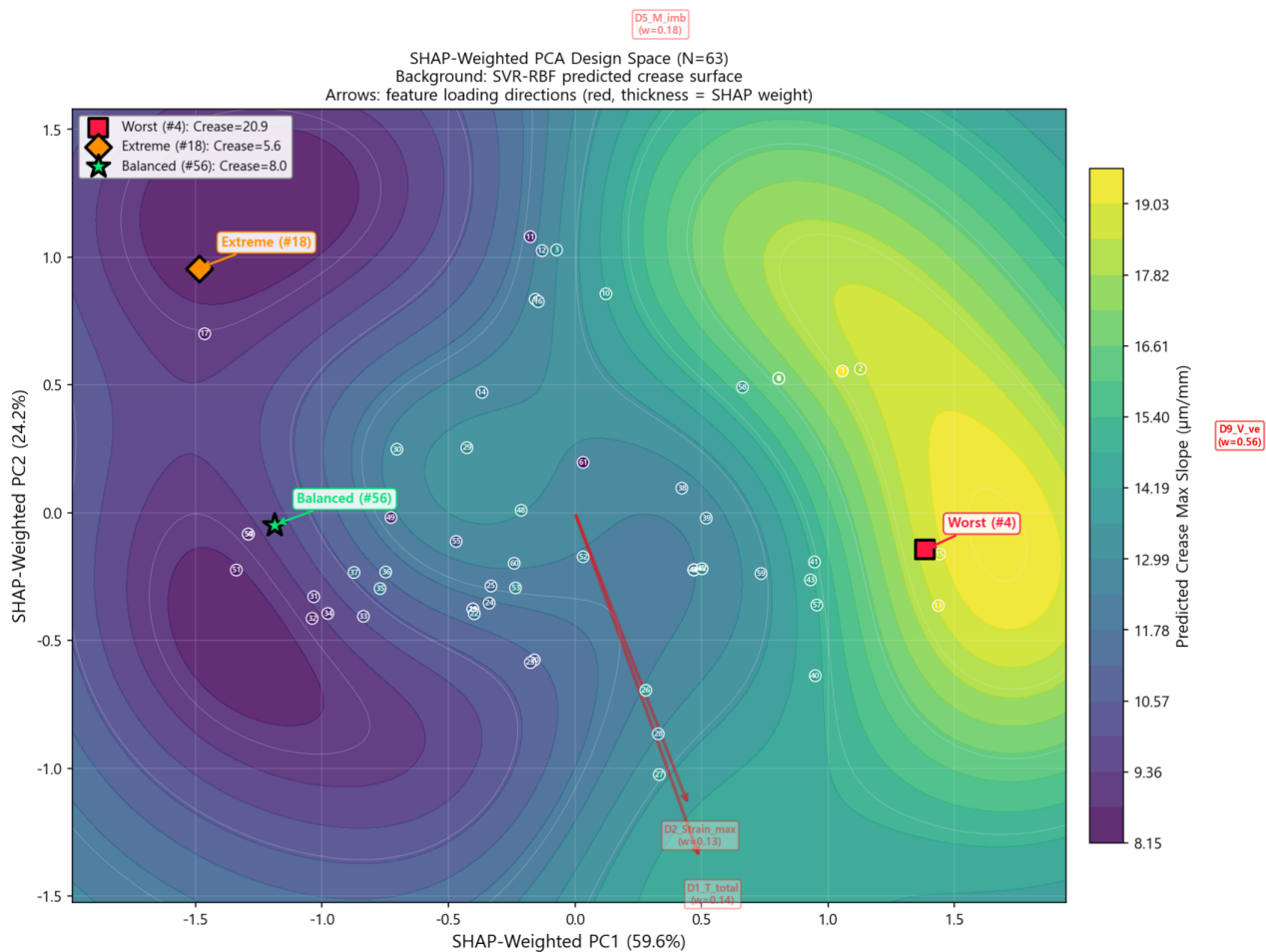


FIGURE 11 | SHAP waterfall comparison: #4 (worst) versus #18 (extreme) versus #56 (balanced design).

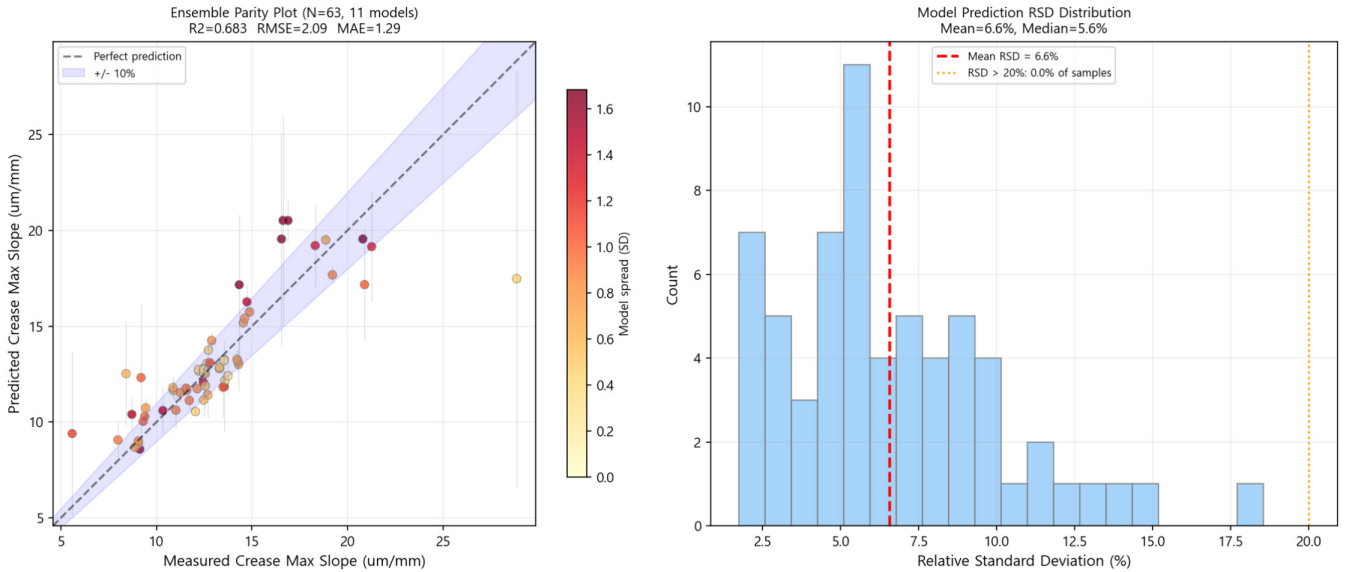


FIGURE 12 | Ensemble parity plot with KNN-LOO error bars and RSD distribution ($N = 63$).

4.10 | Model Reliability and Uncertainty

The ensemble prediction RSD distribution (Figure 12) shows a mean of 6.9% with a median of 5.8%. Only 1.6% of samples exceed an RSD of 20%, indicating robust model consensus. The parity plot with KNN-LOO local error bars ($k = 3$) demonstrates that prediction uncertainty is lowest in the densely sampled mid-crease range and increases toward the extremes (very low or very high crease), consistent with sparser data coverage at the boundaries of the design space.

4.11 | Agent-Analytics Versus Conventional Parameter Optimization

The proposed agent-analytics workflow differs from conventional numerical search approaches such as grid search, genetic algorithms, and Bayesian optimization, as well as from symbolic-regression approaches that search algebraic expressions within a predefined data representation [20, 21]. Although symbolic regression can improve equation-level interpretability, it does not by itself impose the neutral-axis and PAT-based mechanism grouping used here unless such physics constraints are explicitly supplied. These methods typically navigate a predefined numerical design space to maximize an objective function. They may identify a low-crease design candidate, but they do not necessarily explain which physical mechanism makes the design successful or whether the identified parameter combination is transferable across neighboring design regimes.

In contrast, the LLM-assisted component in the present framework operates primarily at the symbolic and mechanistic level. Its role is not to replace the human expert or to autonomously optimize confidential stack parameters, but to help translate qualitative engineering hypotheses into testable mathematical descriptors. For example, hypotheses concerning neutral-axis shift, stiffness-balance asymmetry, viscoelastic accumulation, and interface mismatch were formalized into a structured descriptor family and then evaluated through

ensemble modeling, SHAP stability analysis, and progressive adjustment.

This distinction is central to the proposed framework. A conventional optimizer can search for a numerical optimum within a fixed representation; the agent-analytics workflow helps construct the representation itself. The subsequent evidence hierarchy then separates independently associated mechanisms from pathway-mediated or insufficient descriptors. Thus, the agent contribution is best understood as human-supervised symbolic feature engineering and evidence organization, rather than black-box numerical optimization.

5 | Conclusions

We present an agent-analytics paradigm that converts physics hypotheses into analyzable descriptors and systematically validates them through a three-stage evidence hierarchy. Starting from 14 candidate descriptors organized into five mechanism groups, the framework winnows candidates through ensemble SHAP ranking, bootstrap pathway analysis under progressive adjustment, and progressive confounder adjustment to identify three independently robust drivers: viscoelastic accumulation (D9), dynamic stiffness imbalance (D5), and total thickness (D1).

This systematic winnowing process demonstrates the framework's capacity not only for hypothesis generation but also for systematic screening through progressive adjustment analysis—11 of the 14 descriptors were found to operate through the three base mechanisms or to lack independent support under full adjustment, an outcome that would not have been revealed by predictive modeling alone.

The evolution from ratio-based stiffness symmetry (S_{sym}) to absolute moment imbalance (M_{imb}) was validated by a qualitative shift in evidence-tier status (Tier 3 $\rightarrow$ Tier 1), providing convergent evidence from three independent methods that both descriptors encode the same stiffness-balance mechanism, whereas

absolute moment imbalance offers the more stable operationalization for the present dataset and adjustment framework.

The comparative SHAP waterfall analysis revealed that the best deployable design achieved 8.0 $\mu\text{m}/\text{mm}$ crease (39% below the dataset mean) not through extreme material optimization—which achieved lower absolute crease (5.6) but violated non-crease constraints—but through geometric compensation via stiffness balance. This supports prescriptive guidance: stack designers should prioritize structural balance as a primary crease mitigation lever, complementary to viscoelastic material tuning, to achieve holistically viable designs rather than optimizing a single metric at the expense of system-level reliability.

5.1 | Limitations

Agentic workflows may be sensitive to prompt and tooling choices. We mitigate this via fixed descriptor equations, multi-model ensembles, SHAP stability, and bootstrap adjustment-based screening. Because the present dataset is observational, the results should be interpreted as adjustment-based causal consistency rather than direct interventional proof. The current sample size ($N=63$) limits statistical power for detecting small independent effects; descriptors classified as Tier 4 may exhibit significant effects in larger datasets. Future work will extend the framework to prospective predict–fabricate–measure closed loops and explore causal discovery with interventional data.

Author Contributions

Kyongtae Park: formal analysis, methodology, software, writing – original draft. **Euihoon Hwang:** conceptualization, data curation. **Yeol Choi:** validation, writing – review and editing. **Joo-Hyung Lee:** project administration, writing – review and editing.

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Data Availability Statement

Research data are not shared.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Complete evidence summary for 14 physics-informed descriptors. **Table S2:** Bootstrap ATE

under progressive confounder adjustment. **Table S3:** Parsimonious model performance (Tier 1 + 2 features only). *Note:* The model count varies by analysis stage (up to 14 for ensemble CV, fewer for SHAP computation where kernel-based explanation is not feasible for all model types). **Figure S1:** SHAP dependence for D9 (V_{ve}). Strong monotonic positive trend confirms that higher viscoelastic accumulation consistently increases crease. Color (D5, M_{imb}): High stiffness imbalance (red) amplifies the V_{ve} effect. **Figure S2:** SHAP dependence for D5 (M_{imb}). Threshold behavior: Below ~ 1500 , contributions cluster near -1 to 0 ; above ~ 2000 , contributions shift to $+1$ to $+2$. Color (D1, T_{total}): Thicker stacks (red) show higher M_{imb} sensitivity. **Figure S3:** SHAP dependence for D1 (T_{total}). Monotonic positive trend with saturation above ~ 450 . Color (D2, ϵ_{max}) tracks with T_{total} , confirming that D2 operates through the geometric pathway (Tier 2). **Table S4:** Candidate normalizations considered during S_{sym} to M_{imb} refinement. **Figure S4:** Visual representation of the human-agent dialogue underlying the $S_{sym} \rightarrow M_{imb}$ descriptor refinement, corresponding to Box 1 in the main text. The human expert first frames the physical question in terms of residual stiffness-balance imbalance around the neutral axis (Turn 1, purple). Under this human-framed physical question, the agent generates four candidate normalizations with mechanistic rationales (Turn 2, teal); the rationale shown corresponds to the selected option. The human expert then selects option (c) on physical grounds and specifies empirical verification, including dimensional consistency checking and downstream comparison against S_{sym} using SHAP ranking and progressive-adjustment analysis (Turn 3, purple). The selected formulation yields $M_{imb} = |D_{top} - D_{bot}| / \Sigma D_i$, corresponding to D5 in Section 3.2.4, which achieves Tier 1 status in the evidence hierarchy. Faded boxes indicate candidate formulations not selected. This decision-tree structure illustrates the division of labor in the agent-analytics workflow: human-framed physical reasoning, agent-assisted symbolic candidate generation, and human-verified descriptor selection.